At the same time we cannot say that the sequence

$$1,\ 0,\ 1,\ 0,\ 1,\ 0,\ 1,\ 0,\ 1,\ 0,\ \ldots$$

goes to some number. In other words, this sequence do not converge to any limit. So a sequence may not have a limit. But we can prove that a sequence cannot converge to more than one limit; that is, **if a sequence converges to a limit, then it is unique**.

5.6.1 Fibonacci Sequence



The Fibonacci sequence is a sequence of numbers where a number other than first two terms, is found by adding up the two numbers before it. Starting with 1, the sequence goes 1, 1, 2, 3, 5, 8, 13, 21, 34, and so forth. Written as a rule, the expression is $x_n = x_{n-1} + x_{n-2}, n \geq 3$ with $x_1 = 1, x_2 = 1$

Named after **Fibonacci**, also known as Leonardo of Pisa or Leonardo Pisano, Fibonacci numbers were first introduced in the book Liber abaci in 1202. The son of a Pisan merchant, Fibonacci traveled widely and traded extensively. Mathematics was incredibly important to those in the trading industry, and his passion for numbers was cultivated in his youth.

Knowledge of numbers is said to have first originated in the Hindu-Arabic arithmetic system, which Fibonacci studied while growing up in North Africa. Prior to the publication of Liber abaci, the Latin-speaking world had yet to be introduced to the decimal number system. He wrote many books about geometry, commercial arithmetic and irrational numbers. He also helped in the development of the concept of zero.

n	=	1	2	3	4	5	6	7	8	9	10	11	12	13	14	$\cdots$
x_n	=	1	1	2	3	5	8	13	21	34	55	89	144	233	377	$\cdots$

For example, The 8th term is sum of 6th term and 7th term. Thus, $x_8 = 8 + 13 = 21$.

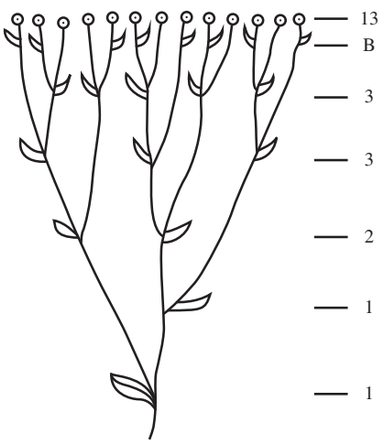


Figure 5.4

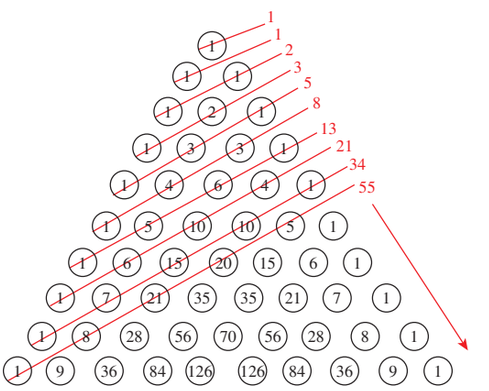


Figure 5.5

There is an interesting pattern in the Fibonacci sequence.

Observe that

(i) every third number is a multiple of 3rd term ($t_3 = 2$).
(ii) every fourth number is a multiple of 4th term ($t_4 = 3$).
(iii) every fifth number is a multiple of 5th term ($t_5 = 5$).
(iv) So, every n^{th} number is a multiple of nth term.